

GOWTHAM PATEL.C

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1. Encrypt "CHASE YOUR DREAMS" using

a) Caesar cipher:

$$\text{Encrypt: } C = E(P+n) \bmod 26$$

$$\text{Decrypt: } P = D(C-n) \bmod 26$$

Key = 3

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
P	Q	R	S	T	U	V	W	X	Y	Z				
15	16	17	18	19	20	21	22	23	24	25				

: Shift 3 letters / position forward

⇒ FKDVH BRXU GUHDPV

b) Shift cipher with key 7:

$$\text{Encrypt: } C = E(P+n) \bmod 26$$

$$\text{Decrypt: } P = D(C-n) \bmod 26$$

Key = 7

: Shift the letters 7 positions forward.

⇒ JOHZL FVBY KYLHTZ

c) ROT 13:

$$\text{Encrypt: } C = E(P+n) \bmod 26$$

$$\text{Decrypt: } P = D(C-n) \bmod 26$$

Key = 13

: Shift the letters 13 positions forward

⇒ PUNFR LBHE QERNZF

2. Decrypt:

a) "WRWDOOB VHFUHW PHVVDJH" using Caesar cipher.

$$\text{Encrypt: } C = E(P+n) \bmod 26$$

$$\text{Decrypt: } P = D(C-n) \bmod 26$$

Key = 3

: Shift letters 3 positions backward

⇒ TOTALLY SECRET MESSAGE

b) "MJQQT BTWQI"

shift cipher 5

$$\text{Encrypt: } C = E(P+n) \bmod 26$$

$$\text{Decrypt: } P = D(C-n) \bmod 26$$

Key = 5

⇒ HELLO WORLD

"SJB DTWP YNRJ"

shift cipher 5

⇒ NEW YORK TIME

\$ "FRAQ ZBER PBSSRR" using ROT13

⇒ SEND MORE COFFEE